DSTL (BCS-303) — Unit 3
Previous Year Questions

ABES Engineering College

Prepared for Practice and Revision

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1 2018–19 (III Semester Theory Exam) Unit 3

Q1: Conditionals & De Morgan

(i) Write the *converse*, *inverse*, and *contrapositive* of: "The home team wins whenever it is raining."

(ii) Show that $\neg(p \lor q)$ is logically equivalent to $(\neg p \land \neg q)$.

Q2: Proving tautology

Show that

$$((p \lor q) \land \neg(q \lor (\neg q \lor \neg r))) \lor (\neg p \land \neg q) \lor (\neg p \lor qr)$$

is a tautology **without using a truth table**.

Q3 (a): Tautology/Contradiction/Contingency & Proof

(i) Definitions (very brief).

- *Tautology:* a formula that is **true for every** truth assignment.
- *Contradiction:* a formula that is **false for every** truth assignment.
- *Contingency:* a formula that is **true for some** and **false for some** truth assignments.

(ii) Show that $(p \lor q) \land (\neg p \lor r) \;\rightarrow\; (q \lor r)$ **is a tautology (no truth table).**

Q3 (b): Argument—Derive *We will be home by sunset*

Premises (translate to symbols). Let

s : It is sunny this afternoon, c : It is colder than yesterday,
w : We will go swimming, t : We will take a canoe trip,
h : We will be home by sunset.

Given:

1. $\neg s \land c$.
2. $w \rightarrow s$ ("We will go swimming only if it is sunny").
3. $\neg w \rightarrow t$.
4. $t \rightarrow h$.

Show: h.

2 2019–20 (III Semester Theory Exam) Unit 3

Q1: Contrapositive of an Implication

Write the contrapositive of the statement: *"If it is Sunday then it is a holiday."*

Q2: Logical Equivalence

Show that the propositions $p \to q$ and $\neg p \vee q$ are logically equivalent.

Q3: Prove a formula is a tautology (no truth table)

Show by *equivalence laws* that

$$\big((P \vee Q) \wedge \neg(\neg Q \vee \neg R)\big) \ \vee \ (\neg P \vee \neg Q) \ \vee \ (\neg P \vee \neg R)$$

is a tautology.

Q4: Principal DNF and CNF

Obtain the **principal disjunctive normal form (PDNF)** and the **principal conjunctive normal form (PCNF)** of $F = (\neg p \to r) \wedge (p \leftrightarrow q)$

Q5: Explain various Rules of Inference for Propositional Logic

State the standard rules of inference (with short examples) used to derive valid conclusions from premises.

Q6: Validity of an Argument

Prove the validity of the following argument:
If the races are fixed, so the casinos are crooked, then the tourist trade will decline. If the tourist trade decreases, then the police will be happy. The police force is never happy. Therefore, the races are not fixed.

3 2020–21 (III Semester Theory Exam) Unit 3

Q1: Negation of an Implication

Write the negation of the following statement: *"If I wake up early in the morning, then I will be healthy."*

Q2: Symbolic Representation

Express the following statement in symbolic form: *"All flowers are beautiful."*

Q3: Validity of Argument

Prove the validity of the following argument:
"If I get the job and work hard, then I will get promoted, then I will be happy. I will not be happy. Therefore, either I will not get the job, or I will not work hard."

Q4

(a) Define tautology, contradiction and contingency. Check whether

$$(p \lor q) \land (\neg p \lor r) \ \rightarrow \ (q \lor r)$$

is a tautology, contradiction or contingency.

(b) Translate the following statements into symbolic form:

(i) The sum of two positive integers is always positive.

(ii) Everyone is loved by someone.

(iii) Some people are not admired by everyone.

(iv) If a person is female and is a parent, then this person is someone's mother.

4 2021–22 (III Semester Theory Exam) Unit 3

Q1: Forms of a Conditional

Translate the conditional statement *"If it rains, then I will stay at home."* into its **contrapositive**, **converse**, and **inverse**. (Let p: "It rains", q: "I will stay at home.")

Q2: Universal Modus Ponens & Modus Tollens

State **Universal Modus Ponens** and **Universal Modus Tollens**.

Q3: Rules of Inference & Consistency

(a) Use rules of inference to justify that the three hypotheses (i) "If it does not rain or if it is not foggy, then the sailing race will be held and the lifesaving demonstration will go on." (ii) "If the sailing race is held then the trophy will be awarded." (iii) "The trophy was not awarded." imply the conclusion (iv) "It rained."

(b) Justify that the following premises are inconsistent. (i) If Nirmala misses many classes through illness then he fails high school. (ii) If Nirmala fails high school, then he is uneducated. (iii) If Nirmala reads a lot of books then he is not uneducated. (iv) Nirmala misses many classes through illness and reads a lot of books.

5 2022–23 (III Semester Theory Exam) Unit 3

Q1: Classify $(p \wedge q) \rightarrow (p \vee q)$

Identify whether $(p \wedge q) \rightarrow (p \vee q)$ is a tautology or a contradiction (use a truth table).

Q2(i): Converse, Inverse, Contrapositive

Express the **converse**, **inverse**, and **contrapositive** of "If $x + 5 = 8$ then $x = 3$."

Q2(ii): Equivalence of $P \leftrightarrow Q$ and $(P \wedge Q) \vee (\neg P \wedge \neg Q)$

Show that $P \leftrightarrow Q$ is equivalent to $(P \wedge Q) \vee (\neg P \wedge \neg Q)$.

Q3(i): Prove validity using rules of inference

Argument:

- If Mary runs for office, she will be elected.
- If Mary attends the meeting, she will run for office.
- Either Mary will attend the meeting or she will go to India.
- Mary cannot go to India.

Conclusion: Thus Mary will be elected.

Q3(ii): Translate into predicate logic

Translate each English statement into symbolic form.

1. For every number there is a number greater than that number.
2. Sum of every two integers is an integer.
3. Not every man is perfect.
4. There is no student in the class who knows Spanish and German.